CASE 21
ATAXIA-TELANGIECTASIA

A defect in double-stranded DNA repair.

Throughout life, each individual's genomic DNA is subject to damage by ionizing radiation, ultraviolet radiation, and reactive oxygen species. These agents cause double-strand breaks in DNA, which are repaired by one of two main mechanisms: nonhomologous end joining (NHEJ) or homologous recombination repair (Fig. 21.1). NHEJ can join DNA ends with little sequence homology to each other and can occur during the G_0, G_1, and G_2-M phases of the cell cycle. In contrast, homologous recombination repair uses a sister chromatid as a template for repairing the broken DNA. Because of the need for a sister chromatid, homologous recombination repair occurs only during the S and G_2 phases of the cell cycle, after a chromosome has been replicated. Some of the same DNA repair proteins are used in both NHEJ and homologous recombination repair.

In many, but not all, instances, repair of DNA double-strand breaks begins with the recruitment of a protein complex, the MRE11:RAD50:NBS1 (MRN) complex, to the site of the break. The MRN complex facilitates the accumulation of the additional proteins necessary for DNA repair. One of the first proteins recruited by the MRN complex is the ATM (ataxia-telangiectasia mutated) protein, which is a serine/threonine kinase. ATM exists as an inactive dimer constitutively bound to a phosphatase that negatively regulates ATM kinase activity. When ATM binds to double-strand DNA breaks, the phosphatase is released. ATM then undergoes autophosphorylation and dissociation into active monomers (Fig. 21.2).

On activation, ATM phosphorylates a large number of downstream proteins involved in DNA repair. Activation of these ATM substrates results in cell-cycle

TOPICS BEARING ON THIS CASE:

- V(D)J recombination
- Class-switch recombination
- Lymphocyte development

This case was prepared by Raif Geha, MD, in collaboration with Janet Chou, MD.

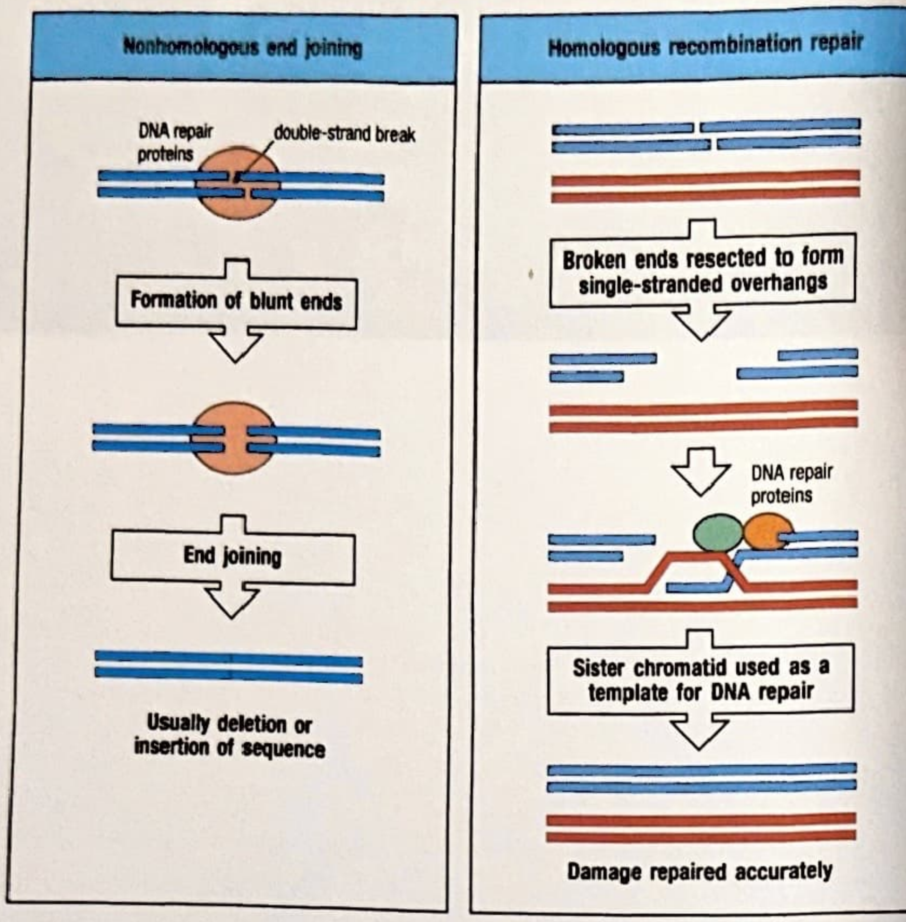

Fig. 21.1 Two methods of repairing double-strand DNA breaks. In nonhomologous end joining (NHEJ) (left), a complex of DNA repair proteins binds to the broken DNA ends. The broken ends are first processed by nucleases and DNA polymerases to remove or fill in single-stranded overhangs, and the blunt ends are then ligated. NHEJ usually leads to the deletion or insertion of nucleotides at the site of the break. An NHEJ-type repair process is used in the ligation of gene segments in V(D)J recombination and in class-switch recombination. In DNA repair by homologous recombination (right), nucleases recruited by a complex of DNA repair proteins transform the broken ends into longer single-stranded overhangs. One strand invades the DNA of the sister chromatid, which is used as a template to repair the DNA break. Unlike NHEJ, homologous recombination uses sequence homology to repair the DNA.

arrest and DNA repair, or in apoptosis if the DNA cannot be repaired. Redundancy in cellular DNA repair mechanisms permits some DNA repair to occur even in the complete absence of ATM. However, cells lacking ATM do not repair every break, and so double-strand breaks accumulate over time, which is manifested as an abnormal sensitivity of cells to ionizing radiation and the accumulation of chromosomal abnormalities.

In addition to its use as a general means of DNA repair, some components of the NHEJ pathway are involved in the rejoining of gene segments to create the signal and coding joints during V(D)J recombination in the immunoglobulin and T-cell receptor genes. Consistent with this, mutations in the genes encoding for Artemis, DNA protein kinase catalytic subunit (DNA-PKcs), DNA ligase 4 (LIG4), and Cernunnos/XLF are responsible for combined immune deficiency with impaired development of T and B lymphocytes (see Case 7). NHEJ is also thought to be the means by which the double-strand breaks produced in the immunoglobulin C-region genes during class-switch recombination are resolved. Both ATM-dependent and ATM-independent pathways operate in these repair processes. As we will see in this case, a genetic deficiency of ATM can result in a constellation of symptoms resulting from the inability to repair DNA, including immunodeficiency.

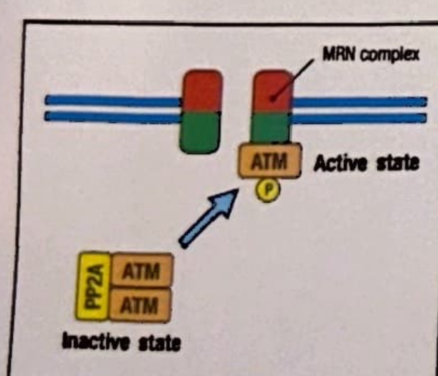

Fig. 21.2 Activation of ATM. The MRN complex binds to double-strand DNA breaks and recruits the protein kinase ATM, which exists in the cytoplasm as an inactive dimer complexed with an inhibitory phosphatase PP2A. ATM activation involves the release of PP2A, which enables the autophosphorylation of ATM and its dissociation into enzymatically active monomers.

The case of Basil Ransom: recurrent infections with difficulty in walking.

Basil Ransom was 6 years old when he was referred to the Children's Hospital because of a history of recurrent ear infections and two pneumonias. Even after placement of ear tubes to ventilate the middle ear and prevent the accumulation of fluid and infections, he continued to have monthly infections. On physical examination he was noted to have a wobbly gait (ataxia) and two dilated superficial blood

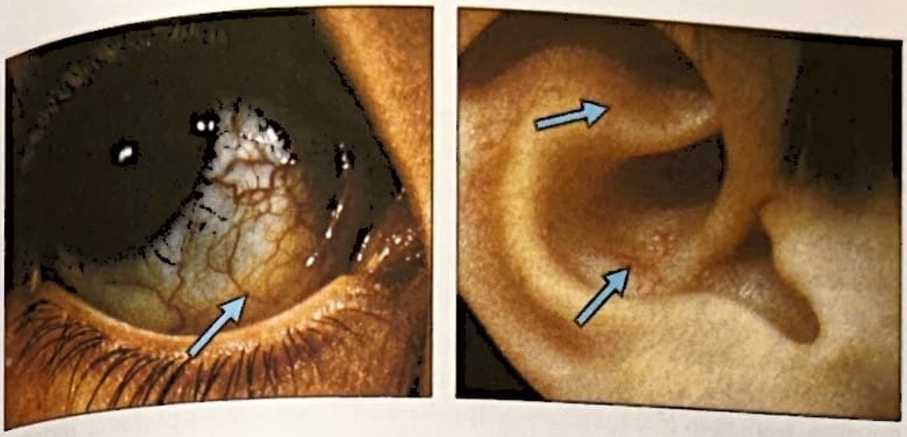

Fig. 21.3 Telangiectasias on the conjunctiva and pinna of the ear in a patient with ataxia telangiectasia.

vessels (telangiectasias) on his conjunctivae and the outer part of his ear (the pinna) (Fig. 21.3). The pediatrician tested Basil's serum immunoglobulins and found total IgG to be low at 219 mg dl^{-1} (normal 600–1500 mg dl^{-1}) with no detectable IgG2 and IgG4, no detectable IgA, and an elevated level of IgM of 205 mg dl^{-1} (normal 22–100 mg dl^{-1}). Basil had a positive antibody titer to only 1 of 14 pneumococcal (*Streptococcus pneumoniae*) subtypes tested, and no detectable antibodies against *Haemophilus influenzae*, although he had received both the pneumococcal and *H. influenzae* vaccines. His antibody titer to tetanus was normal.

Recurrent infections and telangiectasias on conjunctivae and ears.

At the Children's Hospital, a fluorescence-activated cell sorting (FACS) analysis of Basil's peripheral blood mononuclear cells (PBMCs) was notable for low numbers of $CD4^+$ T cells (459 cells μl^{-1}; normal 700–2200 μl^{-1}), $CD8^+$ T cells (221 cells μl^{-1}; normal 490–1300 μl^{-1}), and B cells (300 cells μl^{-1}; normal 390–1400 μl^{-1}). Proliferation of Basil's PBMCs in response to phytohemagglutinin was reduced, but not absent (32,000 counts per minute; normal values 125,000–290,000 cpm). Because of the history of recurrent infections, ataxia, telangiectasias, hypogammaglobulinemia, and lymphopenia, the diagnosis of ataxia-telangiectasia was considered and blood alpha-fetoprotein (AFP) was measured. AFP is a plasma protein produced by the liver that is elevated in ataxia-telangiectasia, although not exclusively so. Basil's AFP level was elevated, at 64 ng ml^{-1} (normal 0–15 ng ml^{-1}).

Undetectable serum IgA. Low $CD4^+$ T cells. Poor antibody responses.

To assess radiosensitivity, lymphoblastoid B-cell lines were established from Basil's blood and from a normal individual as a control. Peripheral blood lymphocytes were isolated and transformed with Epstein–Barr virus. For each sample, the transformed lymphoblastoid cells were split into two plates; one plate was exposed to 10 Grays of radiation and the other was not irradiated. After 10 days in culture, the number of surviving colonies was counted. The survival fraction was calculated by comparing the number of surviving colonies in the irradiated plate with that of the nonirradiated plate. The normal control had a survival fraction of 45%, whereas Basil's survival fraction was only 12% (Fig. 21.4). Sequencing of Basil's *ATM* gene revealed that he was a compound heterozygote for two different mutations; that is, he had one mutation in one *ATM* allele and a different mutation in the other allele.

Radiosensitivity of EBV-transformed B-cell line. Mutation in the ATM gene.

Ataxia-telangiectasia.

Ataxia-telangiectasia is an autosomal recessive disorder characterized by progressive cerebellar ataxia and neurodegeneration, oculocutaneous telangiectasias, primary immunodeficiency, and sensitivity to ionizing radiation. It is caused by mutations in the *ATM* gene. These mutations are typically splice-site or truncating mutations that result in nonproduction of ATM protein. Less commonly, leaky splicing mutations or defects in the promoter region result in low-level expression of functional ATM protein, resulting in a milder form of ataxia-telangiectasia. Most patients, like Basil, are compound heterozygotes for two different *ATM* mutations.

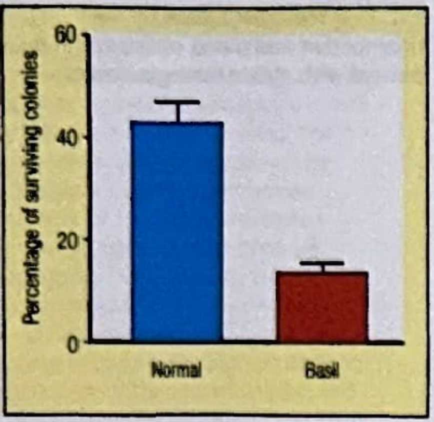

Fig. 21.4 **Radiosensitivity assay of lymphoblastoid cell lines from Basil and a normal control.** The histogram shows the percentage of lymphocytes surviving after irradiation (the survival fraction). The survival fraction for each sample is calculated from the number of surviving colonies on the irradiated plate divided by the number of colonies on a nonirradiated plate.

Ataxia is the earliest sign of the condition, usually beginning at 2–3 years of age. Although the cerebellum is affected first and most severely, the accumulation of damaged DNA in neuronal cells becomes more widespread and results in a loss of gross and fine motor skills, dysphagia (difficulty in swallowing), dysarthria (speech difficulty), abnormal eye movements, and peripheral neuropathy and nerve dysfunction. Although ataxia is associated with several neurologic disorders, the combination of ataxia with telangiectasias on the conjunctivae or pinnae is characteristic of ataxia-telangiectasia. Another identifying feature of the disease is the elevated AFP level, which is usually more than two standard deviations above normal.

Patients with ataxia-telangiectasia can have a variety of defects in humoral and cellular immunity. *ATM* mutations may affect B-cell and T-cell development and function, leading to low numbers of B cells and T cells, most probably owing to deleterious effects on V(D)J recombination. In some cases, T cell lymphopenia is present at birth, evident by abnormally low levels of T cell receptor excision circles (TRECs) at newborn screening for SCID (see Case 5). As a result of *ATM* mutations that impair class-switch recombination, patients may have elevated IgM levels but decreased IgA, IgG, or IgE levels and a poor response to polysaccharide antigens. Consequently, these patients experience frequent infections of sinuses and lungs (sinopulmonary infections) by bacteria such as *S. pneumoniae* and *H. influenzae*. Lymphocyte proliferation in response to mitogens and antigens can vary from normal to impaired. However, unlike patients with ionizing radiation-sensitive severe combined immunodeficiency (IR-SCID) resulting from a deficiency of NHEJ components that are essential for V(D)J recombination, patients with ataxia-telangiectasia are not typically at increased risk of opportunistic infections, indicating the preservation of some immune function, especially T-cell function.

Patients with ataxia-telangiectasia have a significantly increased risk of leukemias and lymphomas. As T cells and B cells routinely incur DNA double-strand breaks during antigen-receptor gene formation and class-switch recombination, the translocations most commonly found in these leukemias and lymphomas involve T-cell receptor genes and immunoglobulin genes on the one hand and oncogenes on the other. Translocations juxtaposing an active T-cell enhancer in the *TCR* locus with an oncogene offer a growth advantage to these abnormal cells, resulting in clonal expansion. Although lymphoma and leukemia constitute the majority of cancers in patients with ataxia-telangiectasia, there is also an increased risk of solid organ cancers, such as breast cancer. The defective repair of double-strand breaks caused by ionizing radiation is the primary cause of tumors in those patients.

Mutations in proteins that cooperate with ATM cause clinical phenotypes that overlap with ataxia-telangiectasia. Hypomorphic mutations (mutations causing decreased, but not absent, production and/or function of the protein) in the *MRE11* gene, which encodes a component of the MRN complex, result in a disease resembling a mild form of ataxia-telangiectasia with delayed onset and slower progression of symptoms. Another component of the MRN complex, nibrin, binds to double-strand DNA breaks. Mutations in the gene encoding nibrin, *NBS1*, cause Nijmegen breakage syndrome (NBS), which is characterized by radiosensitivity, an increased risk of leukemia and lymphoma, and immunodeficiency. Patients with NBS also have microcephaly, mental retardation, short stature, and facial dysmorphisms (long and beaked nose, prominent midface, sloping forehead, micrognathia, and large ears), but do not have ataxia (Fig. 21.5).

Fig. 21.5 **Patient with Nijmegen breakage syndrome showing the characteristic microcephaly, beaked nose, and large ears.** Photograph courtesy of Dr Barbara Pietrucha.

Medical interventions for ataxia-telangiectasia are supportive, because no targeted treatment exists for this disease. The median age of death is 25 years, caused most often by cancer or progressive pulmonary disease resulting from recurrent infections. Therefore, antibiotic prophylaxis should be considered in patients with repeated sinopulmonary infections; pulmonary function should be monitored. Infusions of gamma globulin should be administered to patients with hypogammaglobulinemia or impaired specific antibody production. Most importantly, exposure to ionizing radiation in diagnostic studies should be minimized.

Questions.

1 *Although patients with ataxia-telangiectasia do not suffer from opportunistic infections, there are patients with a syndrome of opportunistic infections, radiosensitivity, and absent or very low numbers of T cells and B cells. What gene defects could cause this clinical picture?*

2 *Most patients with complete ATM deficiency have the classical clinical presentation of ataxia-telangiectasia. However, there are rare patients who lack the ATM protein but have a much milder phenotype with delayed onset of symptoms, mild ataxia, and no history of sinopulmonary infections. What could be a possible explanation for this variability in clinical presentation?*

3 *Why do patients with complete ATM deficiency have a gradual onset of symptoms?*

4 *Radiosensitivity testing can be slow because of the need to establish a lymphoblastoid cell line and to culture the cells. Western blotting for ATM protein is a faster diagnostic tool, because most mutations causing ataxia-telangiectasia are associated with complete absence of the ATM protein. If a Western blot of peripheral blood lymphocytes shows no ATM protein, does this indicate that the patient has ataxia-telangiectasia, even in the absence of clinical symptoms?*